

**Pearson BTEC Level 3 Nationals Certificate, Extended Certificate,
Foundation Diploma, Diploma, Extended Diploma**

Please check the examination details below before entering your candidate information

Candidate surname

Other names

Centre Number

Learner Registration Number

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Time 40 minutes

**Paper
reference**

31617H/1C

**Applied Science / Forensic and Criminal
Investigation**

UNIT 1: Principles and Applications of Science I

Chemistry

SECTION B: PERIODICITY AND PROPERTIES OF ELEMENTS

You must have:

A calculator and a ruler.

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and learner registration number.
- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*

Information

- The exam comprises three papers worth 30 marks each.
Section A: Structures and functions of cells and tissues (Biology).
Section B: Periodicity and properties of elements (Chemistry).
Section C: Waves in communication (Physics).
- The total mark for this exam is 90.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*
- The periodic table of elements can be found at the back of this paper.

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Answer ALL questions. Write your answers in the spaces provided.

Some questions must be answered with a cross ☐. If you change your mind about an answer, put a line through the box ☐ and then mark your new answer with a cross ☐.

1 Lithium and sodium are metals in group 1 of the periodic table.

(a) (i) A physical property of group 1 metals is that they have a low melting point.

Give **one other** physical property of group 1 metals.

(1)

(ii) Group 1 elements have a metallic structure.

Figure 1 shows an incomplete diagram of a metallic structure.

Complete Figure 1 by adding positive and negative charges to the diagram.

(2)

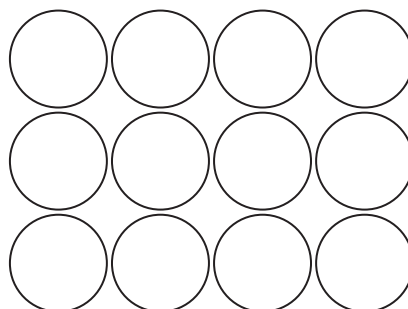


Figure 1

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(b) When a piece of sodium is added to water, sodium hydroxide and a gas are produced.

(i) Identify the gas produced.

(1)

☐ **A** carbon dioxide

☐ **B** hydrogen

☐ **C** nitrogen

☐ **D** oxygen

(ii) Give **one** use of sodium hydroxide.

(1)

(iii) 1.50 g of sodium hydroxide was dissolved in water to make 0.25 dm³ of sodium hydroxide solution.

Calculate the concentration of this solution of sodium hydroxide in g dm⁻³.

(2)

Show your working.

concentration =g dm⁻³



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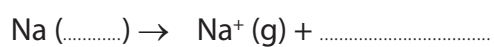
(c) Table 1 shows the first ionisation energies of lithium and sodium.

group 1 element	first ionisation energy (kJ mol ⁻¹)
lithium	520
sodium	496

Table 1

(i) Complete the equation for the first ionisation energy of sodium.

(2)



(ii) Explain why the first ionisation energy of lithium is higher than the first ionisation energy of sodium.

(3)

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(Total for Question 1 = 12 marks)



2 Bromine reacts with iron to form iron(III) bromide.

(a) (i) Identify the structure of iron(III) bromide.

(1)

- ☐ A giant ionic
- ☐ B giant metallic
- ☐ C giant molecular
- ☐ D simple molecular

(ii) Add numbers to balance the equation for the reaction between iron and bromine.

(1)



(b) Bromine and gallium are in period 4 of the periodic table.

The electronegativity of the elements in period 4 changes across the period.

Explain how the electronegativity changes from gallium to bromine.

(4)

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(Total for Question 2 = 6 marks)



3 (a) Potassium is an element in group 1 of the periodic table.

(i) Identify the electronic configuration of a potassium atom.

(1)

- ☐ A $1s^2 2s^2 2p^6 3s^2$
- ☐ B $1s^2 2s^2 2p^6 3s^2 3p^6$
- ☐ C $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$
- ☐ D $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$

(ii) Potassium ions, K^+ , are formed from potassium atoms.

Describe how a potassium atom forms a potassium ion.

(2)

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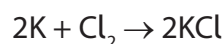
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(b) Potassium chloride can be made by reacting potassium with chlorine.

The equation for the reaction is



Calculate the mass of potassium needed to make 1.492 g of potassium chloride.

(Relative formula mass of potassium chloride = 74.6)

(Relative atomic mass of potassium = 39.1)

(3)

Show your working.

mass of potassium =g

(Total for Question 3 = 6 marks)



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object	observations
iron nail	brown coating, solution loses blue colour
copper coin	no coating, solution remains blue

Table 2

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(6)

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(Total for Question 4 = 6 marks)

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TOTAL FOR SECTION B = 30 MARKS

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The Periodic Table of Elements

1	2	3	4	5	6	7	0 (8)
							(18)
6.9 Li lithium 3	9.0 Be beryllium 4	10.8 B boron 5	12.0 C carbon 6	14.0 N nitrogen 7	16.0 O oxygen 8	19.0 F fluorine 9	20.2 Ne neon 10
23.0 Na sodium 11	24.3 Mg magnesium 12	27.0 Al aluminium 13	28.1 Si silicon 14	31.0 P phosphorus 15	32.1 S sulfur 16	35.5 Cl chlorine 17	39.9 Ar argon 18
39.1 K potassium 19	40.1 Ca calcium 20	69.7 Ga gallium 31	72.6 Ge germanium 32	74.9 As arsenic 33	79.0 Se selenium 34	79.9 Br bromine 35	83.8 Kr krypton 36
85.5 Rb rubidium 37	87.6 Sr strontium 38	114.8 In indium 49	118.7 Sn tin 50	121.8 Sb antimony 51	127.6 Te tellurium 52	126.9 I iodine 53	131.3 Xe xenon 54
132.9 Cs caesium 55	137.3 Ba barium 56	204.4 Tl thallium 81	207.2 Pb lead 82	209.0 Bi bismuth 83	[209] Po polonium 84	[210] At astatine 85	[222] Rn radon 86
[223] Fr francium 87	[226] Ra radium 88	Elements with atomic numbers 112-116 have been reported but not fully authenticated					
		(13)	(14)	(15)	(16)	(17)	
		(12)	(11)	(10)	(9)	(8)	(7)
		(6)	(5)	(4)	(3)	(2)	(1)
Key relative atomic mass atomic symbol name atomic (proton) number							